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1 Introduction

Being able to forecast the federal funds rate well opens the door to smarter policymaking, sharper economic analysis, and for financial markets, it could enable greater stability, better investment strategies, and improved risk management, helping many economic agents navigate uncertainty with greater confidence. Forecasting its movements has therefore been a longstanding concern for policymakers and researchers alike. One of the most prominent frameworks for understanding the federal funds rate is the Taylor rule, which relates the policy rate to deviations of inflation and output from their respective targets. However, while the Taylor rule has proved influential, it faces limitations, particularly its reliance on expected inflation and estimates of potential GDP, which are often unobservable in real time and prone to measurement errors. Its estimation could be a concern as well, since it could be an unbalanced equation.

This paper proposes an alternative approach to forecasting the federal funds rate, aimed at addressing these limitations. Rather than depending on forecasts and estimates of unobserved variables, we develop models based solely on observable data, specifically inflation rates and GDP levels, to predict the future path of monetary policy. Our focus is on empirical time series models—namely ARIMA, ARDL, and ECM specifications—that can incorporate both short-term dynamics and long-term equilibrium relationships without requiring estimates of expected inflation or potential output.

The motivation for this research is twofold. First, enhancing the accuracy of federal funds rate forecasts has important implications for monetary policy transparency and financial market efficiency. Second, designing a model that removes the need for unobservable or forecasted inputs addresses a critical practical challenge in real-time policymaking and forecasting. By comparing the predictive performance of our models against the Taylor rule benchmark, we aim to demonstrate that it is possible to achieve superior forecasting accuracy while working within a more parsimonious and realistic informational framework. Our findings contribute to the broader literature on monetary policy modeling and provide a practical tool for forecasters seeking robust and accessible approaches to predicting the stance of monetary policy.

2 Literature review

2.1 Cointegration

The phenomenon of cointegration was first discovered by Robert Engle and Clive Granger in 1987, as a solution to the spurious regression problem that arises when dealing with non-stationary time series variables. To understand cointegration, it is essential to understand what spurious regressions are and why they are undesirable. When running a regression with two or more nonstationary (i.e., no constant mean and no constant variance) time series variables, OLS results can be seriously misleading. More specifically, when running OLS in a such a context, t-tests and their respective p-values might indicate that there is a statistically significant relationship between a dependent variable and the regressor, while the true regression coefficient could be zero. Additionally, a model might have a high R-squared, indicating a good model fit, while there is no actual relationship. In the case of a spurious relationship, this relatively high R-squared is accompanied by a low Durbin-Watson test statistic, indicating positive residual autocorrelation. When autocorrelation is present but not taken into consideration and accounted for, the usual inferences are invalid and untrustworthy. (Granger & Newbold, 1974)

In most cases, measures have to be taken to make the data stationary, i.e. take first differences, as it is wrong to regress nonstationary variables for the reasons mentioned above. However, an exemption can be made when nonstationary variables are cointegrated. Cointegration appears when variables that are integrated of the same order, have errors that are a linear combination of just the variables.

2.2 Engle-Granger Method

Having k variables, there could theoretically be up to $k-1$ cointegrating relationships between those variables. There is a method to test for one cointegrating relationship with the unit root tests covered in our course. When both a dependent variable (denoted by y_t) and every independent variable (denoted by the vector x_t) are $I(d)$, and the error of regressing y_t on x_t is $I(d-b)$ y_t and x_t are cointegrated with the order $I(d,b)$. (Dolado, Jenkinson et al., 1990) The stationarity of the error term ε_t can be tested by the Augmented Dicky Fuller test, for which the test statistic is calculated as follows:

$$ADF = \frac{\hat{\gamma}}{SE(\hat{\gamma})}$$

$\hat{\gamma}$ is estimated by estimating the following equation:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \varepsilon_t$$

If the value of the test statistic is larger than the critical value, the null hypothesis of unit root presence is rejected, meaning stationarity. It is worth to mention that under the null the teststatistic does not follow a t-distribution. Critical values provided by Dicky-Fuller have to be consulted instead. If one tests with the ADF test the residual of the above mentioned regression for stationarity, one has to pay attention that the critical values are slightly different from the ones provided by Dicky-Fuller. (Webel, 2011)

Next to the Augmented Dicky Fuller test, the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test can be used to test for a unit root, with the following test statistic:

$$KPSS = \frac{1}{T^2} \sum_{t=1}^T S_t^2 / \hat{\sigma}^2$$

where $S_t = \sum_{i=1}^t \hat{u}_i$

In the case of the KPSS test, if the test statistic value is larger than the critical value, the null hypothesis of no unit root (i.e. stationarity) is rejected and there is nonstationarity. Concluding, if the error term may be $I(0)$ (i.e. no unit root) and both the dependent and independent variables may be $I(1)$, there may be one cointegrating relationship between those, according to the Engle-Granger Method.

2.3 Cointegration for interest rates

In macroeconomics, there are several pairs of related variables that are found to be cointegrated, such as substitute goods and inflation and price differentials. For the specific variables considered in this research, namely interest rates and federal funds rates, there have been several papers published in the past years. For example Hansen, who analyzed interest rate relationships between the U.S. and German rates with different maturities by applying the Johansen cointegration method, allowing for multivariate relationships (contrary to Engle-Granger which suits in a bivariate setting) and more than one cointegrating relationship. He found only significant relationships between countries in the short run, while the long-run was predominantly determined by within-country interest rate dynamics. (Hansen, 1996)

Next to this paper, Fu, Su, et al. investigated the possibility of a cointegrated relationship among 91 short-term and long-term Chinese interest rates (with 14 different maturities), using the time frame 2011-2022. More specifically, all these 91 different rates were found to have a statistically significant cointegrated relationship. (Fu, Su et al., 2024)

2.4 Federal funds rate and Taylor rule

With respect to the specific relationship between the federal funds rate and interest rate that will also be discussed in this article, research has been done in different settings. More specifically, Christensen and Nielsen found a long term relationship between the federal funds rate, interest rates and the unemployment rate over the time period 1988-2002 in the United States. (Christensen & Nielsen, 2003) The federal funds rate is the interest rate charged on fund loans that are held by banks in their accounts at the Federal Reserve. (Wheelock, 2022) Christensen & Nielsen also suggested an adapted equation of the Taylor rule (see below), which is a macroeconomic formula that serves as a monetary policy guideline for short term interest rates, in this specific case the federal funds rate.

$$f_t = \pi_t + \lambda_1 \cdot \tilde{u}_t + \lambda_2 \cdot \tilde{\pi}_t + \kappa_0$$

Here, f_t is the federal funds rate at time t , π_t is the inflation, $\tilde{\pi}_t$ is the inflation deviation from the target inflation, $\tilde{u}_t$ is the deviation of economic activity from a natural level (difference between actual real GDP and potential real GDP). Lastly, κ_0 is the equilibrium target real interest rate. (Christensen & Nielsen, 2003) In our research, we assume a target inflation of 2% as stated by the Federal Reserve for their long term inflation objective, more specifically the inflation deviation from the specified target becomes $\tilde{\pi}_t = \pi_t - 2\%$. As this 2% target is constant and thus we do not obtain it empirically, the reduced equation proposed by Christensen & Nielsen under the same conditions is adopted;

$$f_t = \lambda_1 \cdot \tilde{u}_t + (1 + \lambda_2) \cdot \pi_t + \kappa_1$$

Where $\kappa_1 = \kappa_0 - \lambda_2 \cdot 2\%$

2.5 Error correction model

A convenient extension of the cointegration phenomenon is the error correction model. If two variables are cointegrated and of the same order of integration, a regression equation can be formulated as below;

$$\Delta Y_t = \phi + \delta t + \lambda \varepsilon_{t-1} + \gamma_1 \Delta Y_{t-1} + \dots + \gamma_p \Delta Y_{t-p} + \omega_0 \Delta X_t + \dots + \omega_q \Delta X_{t-q} + e_t$$

Here, ϕ is a drift, δ is a trend, ε_{t-1} is the lagged error term of the cointegrated regression, ΔY_{t-p} are the p lagged differences of Y_t , ΔX_{t-q} are the q lagged differences of X_t and e_t is the error term of the error correction model regression. As the name suggests, ε_{t-1} is the error correction term. Note that λ is smaller than zero

2.6 Clark and West test

Ultimately, the objective is to forecast the federal funds rate for the period spanning 2019 to 2024. To evaluate the accuracy of these forecasts, common metrics such as Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and Mean Absolute Error (MAE) will be employed. However, these metrics alone do not account for statistical significance in forecast performance differences, therefore a formal statistical test for out-of-sample predictive accuracy will be conducted, namely the Clark and West (2007) test. This test is used to compare to simpler benchmark models and is particularly suited for when we want to evaluate whether the added complexity of a model leads to meaningful improvements in forecasting performance, rather than simply overfitting the in-sample data.

Let

$$\varepsilon_{1,t} = y_t - \hat{y}_{1,t} \quad \text{and} \quad \varepsilon_{2,t} = y_t - \hat{y}_{2,t}$$

denote the forecast errors from the benchmark and alternative models, respectively. The adjusted forecast error difference at time $t + 1$ is computed as:

$$\hat{f}_{t+1} = \varepsilon_{1,t+1}^2 - \varepsilon_{2,t+1}^2 + (\hat{y}_{1,t+1} - \hat{y}_{2,t+1})^2$$

The average forecast error difference is:

$$\bar{f} = \frac{1}{P} \sum_{t=1}^P \hat{f}_t$$

where P is the number of forecasts (in this case, 6). The Clark and West test statistic is then defined as:

$$t = \sqrt{P} \cdot \frac{\bar{f}}{\sqrt{\widehat{\text{Var}}(\hat{f}_t - \bar{f})}}$$

Under the null hypothesis $H_0 : \mathbb{E}[\hat{f}_t] \leq 0$, the benchmark model performs at least as well as the alternative. The alternative hypothesis $H_a : \mathbb{E}[\hat{f}_t] > 0$ implies that the more complex model provides superior out-of-sample forecasts. This is a one-sided test, since the interest lies in determining whether additional model complexity leads to better predictive performance. The test statistic approximately follows a Student's t -distribution with $P - 1$ degrees of freedom. A significant result provides evidence that the more complicated model does indeed improve out-of-sample forecasting, whereas a non-significant result (with a p-value near 0.5) suggests that the simpler model is just as effective and likely preferable due to its simpler structure and lower risk of overfitting.

3 Methodology

Our objective is to develop an improved empirical model for forecasting the federal funds rate that outperforms the standard Taylor Rule. Specifically, we aim to address two key limitations of the Taylor Rule framework: its reliance on unobservable inputs such as expected inflation and potential output, and its restrictive specification. To overcome these issues, we construct models that use only observable variables—realized inflation and real GDP as predictors.

We begin by examining the time series properties of the FFR, inflation, and GDP. We test for stationarity using the Augmented Dickey-Fuller test and for cointegration using the Engle-Granger two-step method. We find evidence of a cointegrating relationship among the variables and proceed under the assumption that this provides a valid basis for modeling the FFR using error correction models.

We perform forecasts using models from the class of Autoregressive (AR) models, Autoregressive Distributed Lag (ARDL) models, Error Correction Models (ECM), and AR models applied to differenced data. Within each class, we estimate all admissible model specifications, subject to a maximum lag length. This restriction is imposed primarily due to the limited sample size, which constrains the number of parameters that can be reliably estimated. Computational considerations also play a secondary role in setting this limit.

Each model is used to produce one-step-ahead forecasts of the Federal Funds rate based on a growing window approach, where the estimation sample expands recursively with each forecast iteration. Forecast accuracy is assessed using the root mean squared error (RMSE), and the best-performing specifications are identified. We then apply the Clark and West test for nested models to determine whether our selected models offer statistically significant improvements in forecast performance over the benchmark Taylor Rule.

4 Structural breaks and nonstationarity

We began by applying the Augmented Dickey-Fuller (ADF) to assess whether the original time series for inflation, the federal funds rate, and GDP were stationary. In table 1, we see that all variables are initially nonstationary, as for none of the variables the Augmented Dickey-Fuller test rejects the null hypothesis of nonstationarity. However, after taking the first difference of every variable, the Augmented Dickey-Fuller test indicates stationarity. Hence, we conclude that all the variables are integrated of order 1.

Variable	Test statistic	P-value
Federal funds rate	-2.35	[0.43]
First difference	-5.40	[0.01]
Inflation	-2.41	[0.41]
First difference	-5.56	[0.01]
GDP	-1.22	[0.89]
First difference	-4.03	[0.02]

Table 1: ADF test results for stationarity (variables and their respective first differences)

	Critical value: (three variables and $T = 50$)		
Test Statistic	1%	5%	10%
-3.98	-4.59	-3.92	-3.58

Table 2: Engle-Granger Cointegration test

Although both tests indicated non-stationarity across all three variables, it is possible that these tests falsely reject stationarity due to the presence of structural breaks in the data. To address this possibility, we employed the Bai-Perron procedure to detect potential structural breaks. The number of breaks are selected using the Bayesian Information Criterion (BIC), which penalises model complexity to avoid overfitting. If the ADF test fails to reject the null hypothesis of a unit root in the newly segmented time series, we can confidently uphold the original diagnosis of non-stationarity. Otherwise, it would suggest that structural breaks have influenced the initial results and should be included in further analysis.

Table 3: BIC by Number of Breaks (M)

Variable	0	1	2	3	4	5
Inflation	300.8	294.8	251.7	251.3	255.5	264.6
Federal Funds Rate	327.3	314.5	294.2	288.3	281.7	295.6
GDP	1201	1116	1075	1055	1035	1031

As shown in table 3, the Bai-Perron procedure finds the optimal number of breaks (M) to be $M = 3$ for inflation, $M = 4$ for the federal funds rate and $M = 5$ for GDP. These breaks indicate shifts in the data and may mask stationarity.

Table 4 shows that structural shifts in the inflation and interest rate series that are closely aligned with one another and can be explained by known phenomenon such as the low inflation rates after 1990 (Croushore, 1998).

The inflation and federal funds rate series are plotted in Figure 1, with vertical lines indicating the estimated breakpoints. Similarly, the GDP time series with its five breakpoints is presented in Figure 2.

Table 4: Break Points and Dates		
Variable	M	Breakdates
Inflation	3	1972, 1981 & 1991
Federal Funds Rate	4	1971, 1982, 1994 & 2005
GDP	5	1967, 1977, 1987, 1996 & 2005

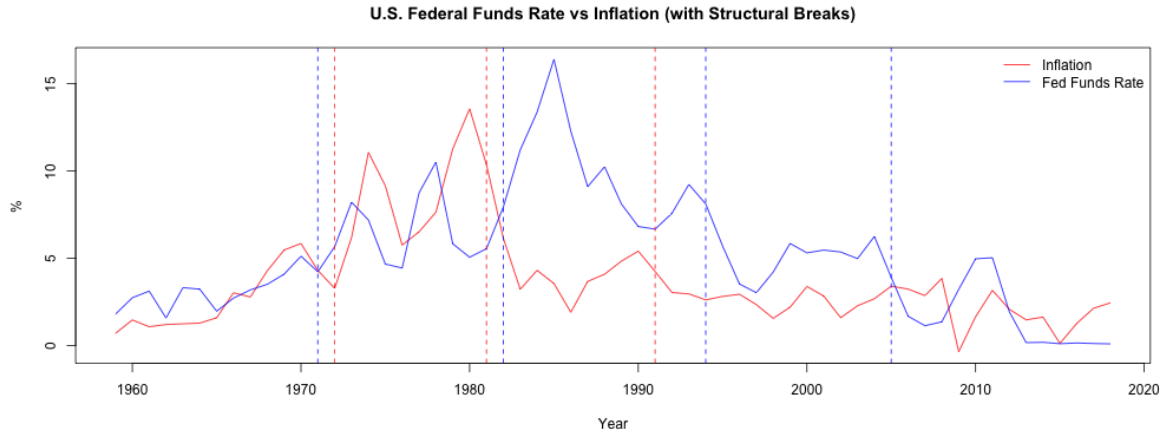


Figure 1:

Figure 2 (bottom) shows a steep and nearly linear upward trend in GDP, reinforcing the non-stationary nature of the level series.

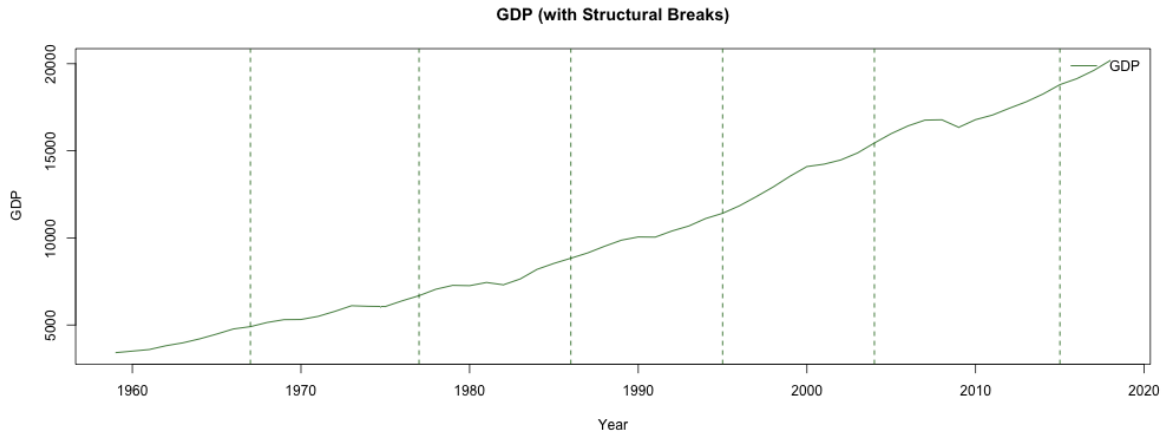


Figure 2:

To assess the potential impact of structural breaks on the stationarity properties of the time series, an Augmented Dickey-Fuller (ADF) test is conducted on each segment identified by the Bai-Perron procedure. The results (Table 4), indicate that non-stationarity persists across most segments of all three series. A single segment of the inflation series (1971–1981) yields a p-value below the 0.05 threshold, suggesting potential stationarity. However the overall pattern remains consistent with the presence of unit roots.

Series	Segment	p-value
Inflation	1956–1970	[0.48]
	1971–1981	[0.05]
	1982–1993	[0.42]
	1994–2004	[0.60]
	2005–2019	[0.91]
Fed Funds	1956–1970	[0.99]
	1971–1981	[0.88]
	1982–1993	[0.87]
	1994–2004	[0.99]
	2005–2019	[0.73]
GDP	1956–1966	[0.99]
	1967–1976	[0.44]
	1977–1986	[0.40]
	1987–1995	[0.97]
	1996–2004	[0.09]
	2005–2019	[0.91]

Table 5: Augmented Dickey-Fuller (ADF) Test Results

These findings support the notion that structural breaks alone do not account for the non-stationarity observed in the original series and therefore we can conclude that the presence of structural breaks was not falsely omitted from the initial analysis. We continue under the assumption that the series remains non-stationary and do not take these potential regime changes into account.

5 Results

5.1 Taylor rule forecast

With respect to the Taylor rule, the model as discussed in the literature review is estimated. The model is estimated with data from 1959 till 2018, and the results are displayed in table 4. Note here that $\tilde{\pi}_t$ is the difference between actual inflation and target inflation (in our case, 2%) and that $\tilde{u}_t$ is estimated as the difference between the logarithm of real GDP and the logarithm of real potential GDP.

Before moving on the the regression results and the forecasts, let us first check the stationarity of $\tilde{u}_t$. From table 4, we see that the variable is stationary, hence it is I(0). In short, the model estimated has an I(1) dependent variable, an I(1) independent variable and an I(0) independent variable.

Test	Test statistic	Conclusion
ADF (With drift)	-4.475	stationary
KPPS	0.088171	stationary

Table 6: $\tilde{u}_t$ stationarity test results

We have to be aware of a possible spurious regression problem or cointegration in this particular case, as the variables used in the regression are not all integrated of the same order. Also, the t-statistics and their respective p-values in table 10 cannot be trusted as usual, as the test statistics do not follow a t-distribution. Note that the adjusted R-squared is relatively high, and that the balance between the relative magnitude of the coefficients (0.1906 versus 0.9981) is quite distorted, where inflation is much more important than the economic output gap. This might indicate a spurious regression problem. However, as our aim is the assess forecasting capability, we will not concern ourselves too much with these in-sample specifications, rather we will see how accurate the predicted values are the model outputs.

To analyze this, the following method is chosen; after the model is trained (with last value 2018), the federal funds rate for 2019 is forecasted. Subsequently, this actual value for 2019 is added to the training data and the value for 2020 is forecasted, and so on till 2024. For the forecasts we use the actual data of that year. This approach can be seen as an expanding window forecast, which we use as the amount of initial training data is limited.

Coefficients	Estimate	Standard error	T-value	P-value
Intercept ($\pi + \kappa_0$)	3.6185	0.4179	8.658	$\ll 0.01$
$\tilde{u}_t$	0.1906	0.1506	1.266	0.211
$\tilde{\pi}_t$	0.9981	0.1049	9.515	$\ll 0.01$
F-statistic	46.21	Adj. R-squared	0.6051	

Table 7: Taylor rule model output

In figure 3, the fitted regression line and the predicted values are visualized. In table 6, the point forecasts generated by the Taylor rule regression are displayed. As seen, the Taylor rule model has limited capability to accurately fit the data. Also, the point forecasts are (except for the last two) off by quite some margin.

Year	Forecasted value	Actual value
2019	5.38	2.16
2020	4.01	0.38
2021	8.21	0.08
2022	11.39	1.68
2023	7.43	5.02
2024	6.34	5.14
RMSE	6.02	

Table 8: Taylor rule point forecasts

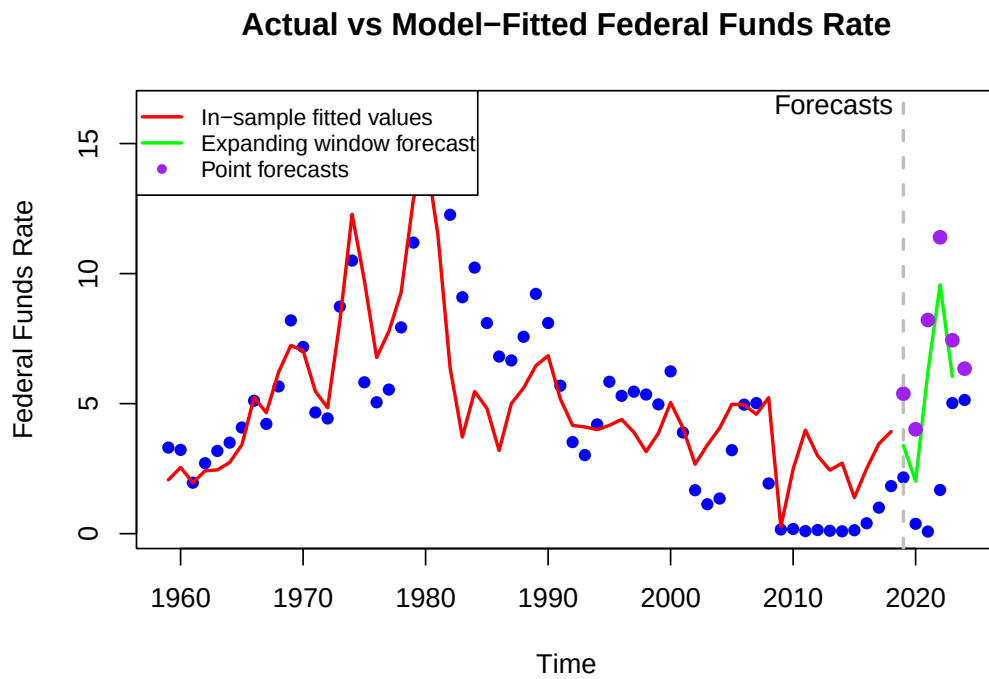


Figure 3: Taylor rule graph

5.2 ECM and ARDL forecasts

The table below presents the five best-performing models in terms of RMSE, drawn from the three model classes under consideration: ARIMA models, ARDL models, and ECM models. For the ECM and ARDL models, the first digit resembles the number of federal funds rate lags, the second digit resembles the number of inflation lags and the third digit resembles the number of GDP lags. To ensure comparability, ARIMA models are restricted to exclude moving average (MA) components, as neither ARDL nor ECM models incorporate MA terms. Among these, the ECM models tend to perform best, achieving the lowest RMSE values overall. ARIMA models follow closely, while the ARDL models lag significantly behind in terms of predictive accuracy.

The strong performance of ECM models is expected, as they explicitly model the cointegrating relationship between the federal funds rate (our dependent variable) and the inflation rate and GDP (our independent variables). By capturing these long-run equilibrium relationships while simultaneously modeling short-run dynamics through the error correction term, ECMs leverage important economic linkages.

The fact that ARIMA models outperform ARDL models further supports the importance of correctly handling non-stationary data. Earlier analysis showed that none of the series involved were stationary. While ARIMA models (except for the AR(5) variant) explicitly account for non-stationarity through differencing, ARDL models do not, which likely explains their poorer predictive performance.

Table 9: Model Comparison based on MAE, MSE, and RMSE (rounded to 2 decimals)

Model	MAE	MSE	RMSE
ECM(1,1,4)	0.48	0.34	0.58
ECM(3,2,5)	0.62	0.46	0.68
ECM(1,2,5)	0.63	0.47	0.68
ARIMA(4,1,0)	0.57	0.53	0.73
ARIMA(5,1,0)	0.57	0.53	0.73
ARIMA(5,0,0)	0.48	0.54	0.74
ECM(4,1,4)	0.61	0.56	0.75
ARIMA(3,1,0)	0.57	0.57	0.75
ARIMA(2,1,0)	0.57	0.57	0.76
ECM(3,1,4)	0.62	0.57	0.76
ARDL(3,1,1)	0.71	0.82	0.90
ARDL(4,1,1)	0.80	0.94	0.97
ARDL(3,1,2)	0.76	0.95	0.97
ARDL(5,1,1)	0.91	1.07	1.03
ARDL(2,1,1)	0.91	1.08	1.04

In order to compare the forecast accuracy of the models against one other and to see whether a each model significantly outperforms others, the Clark and West test is performed. The best performing model according to previous metrics is chosen to be a benchmark: Taylor rule, AR(5), ARIMA(4,1,0), ECM(1,1,4), and ARDL(3,1,1). Since the Taylor rule is a key benchmark, the focus is mostly on comparisons against it.

Table 10: Clark-West p-values comparing each model to selected benchmarks

Model	AR(5)	ARIMA(4,1,0)	ECM(1,1,4)	ARDL(3,1,1)	Taylor Rule
ARIMA(2,1,0)	–	–	–	–	0.05
ARIMA(3,1,0)	–	–	–	–	0.05
ARIMA(4,1,0)	–	–	–	–	0.05
ARIMA(5,1,0)	0.75	0.13	–	0.18	0.05
ARIMA(5,0,0)	–	0.13	–	0.18	0.05
ECM(1,1,4)	0.68	0.63	–	0.18	0.02
ECM(1,2,5)	0.67	0.63	0.17	0.18	0.02
ECM(3,2,5)	0.69	0.65	0.22	0.18	0.02
ECM(3,1,4)	0.66	0.61	0.40	0.18	0.02
ECM(4,1,4)	0.66	0.61	0.40	0.18	0.02
ARDL(2,1,1)	–	0.79	–	–	0.03
ARDL(3,1,1)	0.79	0.79	–	–	0.02
ARDL(3,1,2)	0.86	0.86	0.90	0.88	0.01
ARDL(4,1,1)	0.79	0.79	0.77	0.83	0.03
ARDL(5,1,1)	0.79	0.79	0.77	0.16	0.03

he AR and ARIMA models fail to perform significantly better than the Taylor rule ($\alpha = 0.05$), despite some coming very close. In contrast, several ECM and ARDL models do reject the null, indicating they provide meaningful improvements over the Taylor rule. The AR(5) model is not significantly outperformed by any other model, which is surprising given that it is misspecified due to the lack of differencing in non-stationary data. No model is able to beat out any of the benchmark models, other than the benchmark Taylor rule. Interestingly no model is able to beat out the benchmark model other than the benchmark Taylor rule. This would suggest that the top-performing models are fairly close to each other in terms of forecasting ability. It could hint at the fact that all the models still struggle to deliver highly accurate forecasts and it could also suggest that the out-of-sample window of six observations is quite small and could limit the power of the test. While the ECM models yield the lowest RMSE values, they do not significantly outperform the other strong models. At the same time, they are not significantly worse than any alternative either. This suggests they are reliable but not clearly dominant. It's also worth noting that the Clark and West test accounts for potential overfitting. So, while some models offer slightly better in-sample accuracy, the gain may not be large enough to offset the increased risk of overfitting. This may explain why the improvements in forecast accuracy are not always found to be statistically significant.

6 Conclusion

This study investigates the dynamic relationship between the federal funds rate, GDP, and inflation using the Engle-Granger cointegration approach. We identify one counteracting long-run relationship among the variables, which, when modeled within an Error Correction Model (ECM) framework, delivers the most accurate forecasts in terms of RMSE. To benchmark this performance, we compare the ECM against a variety of alternative models—including autoregressive (AR), differenced AR, and ARDL specifications—many of which significantly outperform the Taylor Rule based on the Clark and West test. Nonetheless, while the ECM excels in point forecast accuracy, it does not exhibit a statistically significant improvement over the benchmark models in forecast comparison tests. These results suggest that while incorporating long-run equilibrium relationships can enhance forecast precision, the gains may not always translate into statistically significant improvements over simpler models.

Given the ECM model's superior forecast accuracy in terms of RMSE but lack of statistical significance over benchmarks in the Clark and West test, future research could explore model extensions that incorporate time-varying parameters or regime-switching dynamics to better capture structural shifts in monetary policy or macroeconomic relationships. Additionally, investigating alternative cointegration techniques—such as the Johansen method—or expanding the model to include additional macroeconomic variables (e.g., unemployment, financial conditions indices) may yield further improvements.

7 Discussion

Although we tried to be as precise as possible with the methodology and steps taken in this research, there are some critical notes to be made. Firstly, the data used has yearly frequency, which might impact the results as with yearly frequency we might overlook short term dynamics more than with high frequency of observations. Also, the training data set used was quite small, namely only 59 observations. This might have the implications that our model estimates were too much affected by outliers, as well as the fact that there is a higher risk of overfitting.

Next, with respect to comparison of the models, we have to note that the Taylor rule model and the ECM, ARDL and ARIMA models have different variables that are included. Namely, the Taylor rule model includes the difference of the log of GDP and the log of potential GDP, whereas the ECM, ARDL and ARIMA models use just GDP. In terms of the models, for the Taylor rule this might have had a significant impact because the difference of the log of GDP and log of potential GDP is stationary (and just GDP is not), so that there might have been a spurious regression problem because the federal funds rate and inflation are integrated of order 1, and the difference of the log of GDP and log of potential GDP is integrated of order 1. Although the point of the Taylor rule is not to account for a cointegrating relationship, it might be inaccurate to directly compare the Taylor rule and ECM/ARDL models in terms of forecasting accuracy as they include different variables.

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